

Article

AERIS: Environment-Aware Hierarchical Routing for Reliable Wireless Sensor Networks under Realistic Channel Conditions

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Abstract: This paper studies AERIS, an environment-aware hierarchical routing protocol for wireless sensor networks, with packet delivery ratio (PDR) as the primary outcome. In a legacy 100-node comparability setting (four environments, 30 seeds per environment), AERIS achieves the highest mean PDR among five protocols (AERIS, LEACH, PEGASIS, HEED, and TEEN). In the strict large-scale setting (100–1000 nodes, four environments, $n=3200$ per environment-node-protocol cell, explicit MAC-collision and multi-hop relay modeling), AERIS is rank-1 in three environments, while PEGASIS dominates indoor_office at high scale. A three-level transmit-power sensitivity matrix (5/10/15 dBm, $n=1000$ per cell) shows strong environment-dependent and non-monotonic responses. Matched stress-delta comparisons (patch minus control, $n=1000$ vs 1000) show consistent absolute PDR degradation under stricter physics assumptions, indicating that deployment requires explicit calibration rather than fixed global settings. NS-3 is used for trend-level cross-platform validation for AERIS versus LEACH. The overall evidence positions AERIS as a reliability-oriented option for harsh channels with bounded operating regimes.

Keywords: wireless sensor networks; reliable routing; hierarchical protocol; packet delivery ratio; scalability; reproducible evaluation

1. Introduction

Wireless sensor network (WSN) routing is classically optimized for energy reduction, but practical deployments also require stable delivery under heterogeneous channels [1–3]. This creates a multi-objective tension among reliability, hop-based latency, and energy use.

Existing protocols represent distinct operating points: LEACH emphasizes clustered forwarding [4], PEGASIS emphasizes chain-based aggregation [5], HEED emphasizes residual-energy-aware head selection [6], and TEEN emphasizes event-triggered reporting [7]. Prior channel studies also show that low-power links can be asymmetric and environment-sensitive under realistic propagation conditions [8–10].

AERIS prioritizes reliability under realistic channel variability as a lightweight hierarchical protocol. The design objective is not universal dominance; rather, it is robust performance across multiple channel environments while preserving deployability on resource-constrained nodes.

1.1. Contributions

This manuscript makes three contributions:

- a modular routing design (CAS switching, Skeleton backbone, Gateway reinforcement) with explicit environment-scoped operation boundaries and auditable control variables;

- a regime-separated evidence design that distinguishes core reliability reporting (100-node and primary large-scale settings) from stress and sensitivity diagnostics;
- cross-platform NS-3 trend validation for AERIS versus LEACH under aligned environment taxonomy.

The manuscript emphasizes reproducible conclusions under explicit metric, environment, baseline, and sample-size definitions.

2. Related Work

Classical WSN protocols (LEACH, PEGASIS, HEED, TEEN) provide foundational routing mechanisms but show different failure modes under harsh channels and larger scales [3,5,6,11]. Chain-based aggregation can reduce per-hop transmission effort but increases end-to-end hop count. Cluster-based approaches can reduce hop count but may become sensitive to uplink quality and head placement.

Recent adaptive or learning-based methods improve specific settings but often require heavier runtime or training overhead [12–20]. Recent Sensors/IoT studies report gains from context-aware or hybrid routing under constrained networks [21–29].

Additional recent studies show that routing quality is strongly coupled to link asymmetry, deployment context, and adaptive control-policy design [30–35]. NLOS-aware estimation methods further demonstrate that environment uncertainty degrades localization and, by extension, distance-dependent routing decisions [36]. These results motivate explicit environmental adaptation in protocol design while keeping inference and control logic lightweight.

Hybrid cluster-chain switching and gateway-assisted hierarchical forwarding are not new concepts per se; earlier work explored LEACH-PEGASIS hybrids and energy-efficient hierarchical gateway routing. AERIS differs in two specific respects: (1) CAS is not a learned policy or meta-heuristic but a fixed-coefficient linear scoring rule whose mode weights are shared across all environments, making behavior fully auditable and reproducible without training; (2) the Gateway module operates as a conditional fallback layer (activated only when direct uplink quality is insufficient), not as a permanent relay tier, which avoids single-point-of-failure concentration at gateway nodes.

In this manuscript, the focus is a rule-based protocol with explicit reproducibility constraints and auditable experiment metadata. The baseline set (LEACH, PEGASIS, HEED, TEEN) is selected to span representative WSN routing paradigms (cluster-head rotation, chain forwarding, residual-energy clustering, and threshold-triggered event reporting) under a comparable low-overhead control budget.

3. System Model and Protocol

3.1. Metric Definition

The primary metric is

$$\text{PDR}_{\text{expected}} = \frac{N_{\text{delivered}}}{N_{\text{expected}}}, \quad (1)$$

where $N_{\text{delivered}}$ is the number of unique data packets received at the base station and N_{expected} is the number of packets that *alive* source nodes were scheduled to generate during all rounds (i.e., one packet per alive node per round). This denominator is protocol-independent: it counts expected application-layer transmissions regardless of whether a protocol suppresses sending (as in TEEN's threshold-based reporting). All publication conclusions in this draft are based on this metric.

3.2. AERIS Architecture

AERIS combines three components:

1. Context-Adaptive Switching (CAS) for mode control,
2. Skeleton backbone for hierarchical forwarding,

3. Gateway coordination for CH-to-BS reinforcement.

Gateway candidate scoring follows

$$\text{Score}_i = \alpha \hat{E}_i + \beta \hat{C}_i + \gamma \hat{L}_i + \delta \hat{D}_i, \quad (2)$$

where $\hat{E}_i$ is normalized residual energy, $\hat{C}_i$ is normalized CH centrality, $\hat{L}_i$ is normalized gateway-link quality, and $\hat{D}_i$ is normalized CH-to-BS distance. CAS mode selection is modeled as

$$m^* = \arg \max_{m \in \{\text{direct}, \text{chain}, \text{twohop}\}} \left(w_{1,m} \hat{q}_m + w_{2,m} \hat{e}_m + \mathbf{w}_{3,m}^\top \hat{\mathbf{u}}_m + b_m \right), \quad (3)$$

where $\hat{q}_m$ and $\hat{e}_m$ are normalized link-quality and energy features, and $\hat{\mathbf{u}}_m = [\hat{d}_m, \hat{r}_m, \hat{\rho}_m, \hat{f}_m]^\top$ expands structural uncertainty into normalized distance-to-BS, cluster radius, density, and fairness features. The fixed coefficients used in publication-tier runs are listed in Table 1. In Eq. (2), the negative distance coefficient ($\delta < 0$) penalizes long CH-to-BS distance and therefore biases selection toward candidates with lower expected uplink attenuation.

Table 1. Fixed initialization coefficients for publication-tier runs (shared across all four environments).

Component	Mode	w_q	w_e	w_d	w_r	w_ρ	w_f
CAS	direct	0.65	0.35	-0.25	-0.05	0.10	-0.05
CAS	chain	0.40	0.30	0.20	0.20	0.20	-0.05
CAS	two-hop	0.25	0.20	0.50	0.15	0.05	-0.05
Component		$\alpha (\hat{E})$	$\beta (\hat{C})$	$\gamma (\hat{L})$	$\delta (\hat{D})$		
Gateway		0.15	0.20	0.35	-0.60		

Runtime stage-adaptive updates are bounded, deterministic under fixed seeds, and logged as reproducibility metadata. Source-to-manuscript traceability is provided in the supplementary coefficient mapping material.

3.3. Reliability-Gain Intuition

The key design intuition is conditional path diversity under uncertain links. In the implemented uplink logic, forwarding is a cascaded first-hit policy (direct quality check $\rightarrow$ Gateway route if available $\rightarrow$ Skeleton route if available $\rightarrow$ best-effort direct fallback), not a simultaneous multi-path transmission. Let p_d , p_g , p_s , and p_b denote successful-delivery probabilities for direct, Gateway, Skeleton, and best-effort fallback paths under the same round state. With binary availability indicators $I_g \in \{0, 1\}$ and $I_s \in \{0, 1\}$, an implementation-consistent success expression is

$$P_{\text{succ}} = p_d + (1 - p_d) [I_g p_g + (1 - I_g) I_s p_s + (1 - I_g)(1 - I_s) p_b]. \quad (4)$$

Eq. (4) is therefore a conditional reliability decomposition aligned with the actual control flow, under the assumption of conditionally independent path outcomes given the same round state and availability gates. CAS chooses the lower-cost intra-cluster collection mode before this uplink stage, so routing overhead is bounded while uplink reinforcement remains available when direct quality is insufficient.

3.4. Design Rationale and Decision Logic

The protocol design follows three engineering objectives that are directly linked to known failure modes of classical WSN routing.

- **Reliability under harsh channels:** when direct CH-to-BS links become unstable, Gateway support and Skeleton routing create additional forwarding opportunities instead of relying on

a single uplink path. If the elected gateway node itself fails (energy depletion or link loss), the cascaded fallback logic in Eq. (4) bypasses the gateway layer and attempts Skeleton routing or best-effort direct transmission; gateway re-election occurs at the next round boundary without protocol-level oscillation.

- **Low online overhead:** CAS is a fixed-coefficient rule model with three candidate modes; no online training or iterative optimization is introduced at runtime.
- **Auditable behavior:** all mode choices are deterministic under the same input state, and publication-tier runs fix a shared coefficient set across all four environments.

At each round, AERIS first performs CH election and clustering, then applies CAS mode selection for intra-cluster forwarding, and finally performs CH-to-BS forwarding with optional Gateway/Skeleton reinforcement. This ordering separates local data-collection decisions from long-range forwarding decisions and avoids cross-stage coupling in the control loop.

3.5. Round Execution and Complexity

Each round executes the following fixed sequence (held constant across protocols to preserve comparability):

1. Compute CH scores and elect CH set.
2. Associate non-CH nodes to CHs.
3. For each cluster, evaluate CAS features $(\hat{q}, \hat{e}, \hat{u})$ and select one mode from {direct, chain, twohop}.
4. Aggregate data at CH.
5. If Gateway is enabled, rank gateway candidates with Eq. (2); if Skeleton is enabled, build/update backbone links.
6. Forward CH traffic to BS and log metrics.

Figure 1 summarizes this execution and evidence pipeline.

Let N be the node count and H be the CH count in one round. CH election and clustering are $O(NH)$, CAS selection is $O(1)$ per cluster, and Gateway ranking is $O(H \log H)$. Skeleton update scales with active candidate links E_a . The dominant per-round cost is therefore

$$T_{\text{round}} = O(NH + H \log H + E_a), \quad (5)$$

where $H \ll N$ in the tested regimes. This expression is an implementation-level bound, not a formal optimality claim.

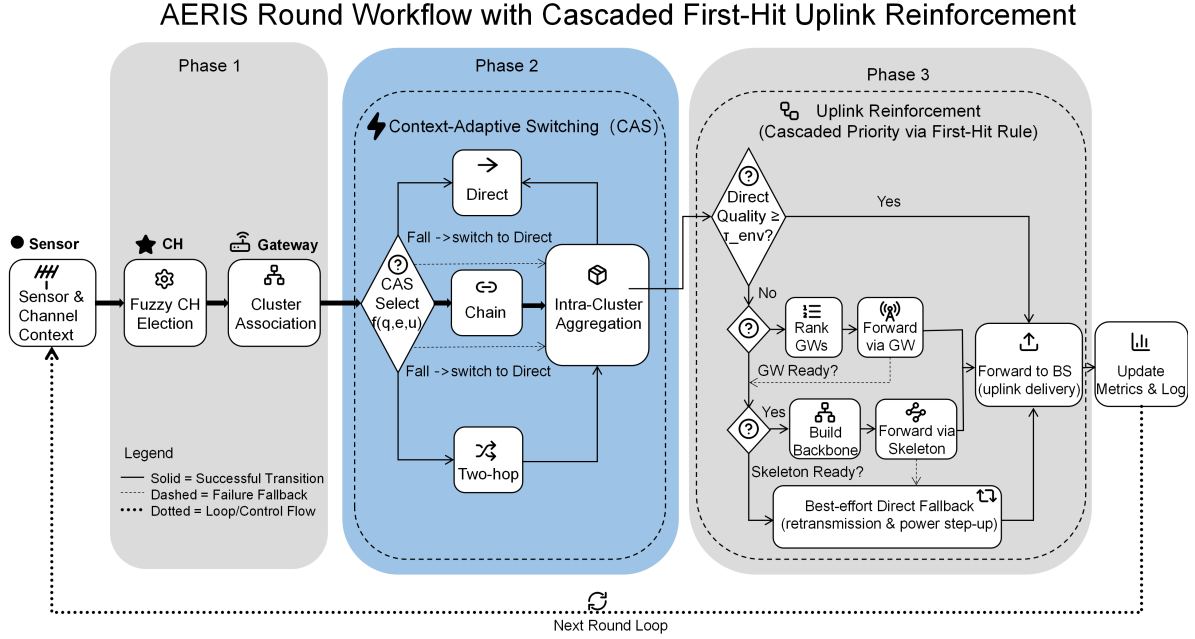


Figure 1. Three-phase AERIS workflow from per-round control decisions to evidence output. The diagram separates routing decisions, safety control, and metric/provenance logging to keep reproducibility boundaries explicit.

4. Experimental Setup

4.1. Publication-Tier Settings

- Simulator: Python implementation with logged provenance.
- Channel environments: indoor_office, indoor_factory, outdoor_urban, outdoor_suburban.
- Default node count: 100.
- Seeds for multi-environment and ablation: 30 independent seeds.
- Seeds for scalability: primary large-scale matrix with $n=3200$ per environment-node-protocol cell (100–1000 nodes, 4 environments; strict-physics flags enabled).
- Baselines: LEACH, PEGASIS, HEED, TEEN.

Channel model. Each environment uses a log-distance path loss model with log-normal shadowing. Received power at distance d is $P_{rx}(d) = P_{tx} - PL(d_0) - 10n \log_{10}(d/d_0) - X_{\sigma}$, where n is the path loss exponent and $X_{\sigma} \sim \mathcal{N}(0, \sigma^2)$ is shadowing. Table 2 lists the per-environment parameters.

Table 2. Channel model parameters per environment.

Environment	Path loss exponent n	Shadowing σ (dB)
indoor_office	2.0	4.5
indoor_factory	2.7	8.5
outdoor_urban	3.4	12.0
outdoor_suburban	2.8	7.5

Energy model. All protocols share the first-order radio model: $E_{tx}(k, d) = E_{elec} \cdot k + \epsilon \cdot k \cdot d^{\alpha}$, $E_{rx}(k) = E_{elec} \cdot k$, with $E_{elec} = 50$ nJ/bit, $\epsilon_{fs} = 10$ pJ/bit/m² (free space, $d < d_0$), $\epsilon_{mp} = 0.0013$ pJ/bit/m⁴ (multipath, $d \geq d_0$), crossover distance $d_0 = \sqrt{\epsilon_{fs}/\epsilon_{mp}} \approx 87$ m, packet size $k = 4000$ bits, and initial energy $E_0 = 0.5$ J per node.

MAC and collision model. In the primary large-scale matrix (strict-physics mode), a slot-level collision flag is enabled: if two or more nodes in the same cluster transmit in the same TDMA slot, all colliding

packets are dropped. Baseline protocols that lack native multi-hop relay (LEACH, HEED, TEEN) are augmented with a simple greedy relay mechanism to ensure fair comparison on multi-hop paths. No capture effect or retransmission is modeled; a collided packet is unconditionally lost.

4.2. Evidence Scope

The analysis is grounded in publication-tier multi-environment comparison, ablation, scalability, hop-based latency, energy-lifetime statistics, and NS-3 trend validation. Cross-regime numerical pooling is avoided.

4.3. Fairness Controls and Simulator Credibility

The simulator is custom Python code, so fairness controls are explicitly enforced in publication-tier runs:

- identical tx-power configuration across protocols in the same matrix,
- identical environment taxonomy and node-scale grid per protocol,
- deterministic seed schedules per regime,
- no reliability override (forced reliable delivery disabled).

To reduce implementation-bias risk, the primary large-scale matrix removes previously identified baseline suppressors and reruns the four-environment scalability matrix under explicit collision and relay settings. NS-3 is then used as an independent trend-level check for AERIS versus LEACH under aligned environment definitions. NS-3 is therefore used as external directional validation rather than as a one-to-one numerical replacement of the Python matrix.

4.4. Simulator Regimes and Physics Assumptions

This study reports two regime tiers for scalability interpretation and keeps the naming consistent across all tables and captions. The primary tier is the large-scale matrix (explicit collision-and-relay settings, balanced $n=3200$ per cell) and is used for large-scale ranking claims. The secondary tier (stress and sensitivity blocks) is retained for matched stress comparisons (patch minus control) and tx-power diagnostics, not for replacing the primary ranking matrix.

For readability, we refer to patch-minus-control contrasts as *stress-delta* comparisons (i.e., the signed difference in PDR between the patch arm with strict physics enabled and the control arm with legacy configuration, under matched environment and node settings) throughout Sections 5.4 and 5.6.

Table 3. Evidence-regime map used in this manuscript.

Block	Scope	Typical sample size	Role in claims
100-node matrix	4 env $\times$ 5 protocols	$n=30$ per env-protocol cell	Core reliability comparison
Primary large-scale matrix	4 env $\times$ 6 node counts $\times$ 5 protocols	$n=3200$ per env-node-protocol cell	Primary large-scale evidence
Stress-delta matrix A (patch minus control)	4 env $\times$ 6 node counts $\times$ 5 protocols	patch $n=1000$, control $n=600$	Physics-stress calibration
Sensitivity block (tx-power matrix)	4 env $\times$ 3 tx levels $\times$ 6 node counts $\times$ 5 protocols	$n=1000$ per env-node-protocol cell	Tx-power boundary analysis
Stress-delta matrix B (matched patch-minus-control)	4 env $\times$ 6 node counts $\times$ 5 protocols	patch $n=1000$, control $n=1000$	Matched strict-physics confirmation
NS-3 alignment	4 env $\times$ 7 node counts, AERIS vs LEACH	$n=30$ per env-node-protocol cell	Trend-level cross-platform check

Table 3 defines the claim boundary used throughout the Results and Discussion sections.

4.5. Statistical Methods

For pairwise protocol comparisons, we use Welch's two-sample t -test:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}, \quad (6)$$

because variance equality is not assumed across protocols and environments. Family-wise error control is handled by Holm correction within each reported comparison family. Each family is defined as the

set of pairwise comparisons within a single table or figure: for example, the 100-node matrix (Table 4) defines one family of $m = 4$ tests (AERIS vs each baseline per environment); the scalability heatmap (Figure 5) defines one family per environment column ($m = 4$ baselines $\times$ 6 node counts = 24); the ablation analysis (Table 5, Figure 3) defines one family of $m = 8$ tests (two ablation variants $\times$ four environments). Effect size is reported with Hedges' g :

$$g = J \cdot \frac{\bar{x}_1 - \bar{x}_2}{s_p}, \quad J = 1 - \frac{3}{4(n_1 + n_2) - 9}, \quad (7)$$

and interpreted as small (0.2), medium (0.5), and large (0.8) in absolute value following Cohen's heuristics; however, because these thresholds are not calibrated for simulation-derived PDR distributions, we primarily use g to convey practical magnitude and always report the accompanying absolute Δ PDR alongside it. Because Welch's t does not assume equal variance but Hedges' g uses pooled s_p , we verified that within-cell coefficient-of-variation ratios remain below 2:1 in publication-tier matrices; when this condition is violated the pooled standardizer would overstate or understate the effect, and we flag such cells as unreliable in the significance maps. Confidence intervals are reported as two-sided 95% intervals. All significance statements in this manuscript are based on Holm-corrected p values. For large- n cells, interpretation is based jointly on corrected p values, effect size, and absolute PDR delta, because with $n = 3200$ even trivially small effects reach statistical significance. Reported table-wise standard deviations are treated as descriptive values from the corresponding result files; inferential conclusions are computed from raw replicate-level records.

4.6. Reproducibility Controls

All publication-tier scripts in this study use deterministic seed lists and disable reliability-override logic. This guarantees that PDR reflects simulated delivery behavior rather than forced delivery assumptions. We treat hop count as a latency proxy, not a wall-clock latency measurement. Significance statements are accepted only when consistent with the declared regime-specific sample size.

4.7. Auditability Constraints

To keep claims machine-checkable, each publication-tier output includes run-tier tags, metric tags, and script/hash sidecar metadata. Claims in the main text are restricted to matrices that satisfy declared sample-size and metric constraints. This prevents mixing exploratory diagnostics with publication-tier statements.

5. Results

5.1. Multi-Environment Comparison at 100 Nodes ($n=30$)

Table 4. Legacy comparability matrix (strict collision/relay flags disabled): PDR by environment at 100 nodes ($n=30$ per environment-protocol cell).

Environment	AERIS	LEACH	PEGASIS	HEED	TEEN
indoor_office	0.9739 $\pm$ 0.0047	0.5543 $\pm$ 0.0401	0.9078 $\pm$ 0.0166	0.9371 $\pm$ 0.0076	0.8222 $\pm$ 0.0044
indoor_factory	0.6031 $\pm$ 0.0258	0.1614 $\pm$ 0.0209	0.1928 $\pm$ 0.0255	0.2326 $\pm$ 0.0263	0.3113 $\pm$ 0.0245
outdoor_urban	0.3745 $\pm$ 0.0354	0.0552 $\pm$ 0.0127	0.0542 $\pm$ 0.0117	0.0635 $\pm$ 0.0121	0.1201 $\pm$ 0.0183
outdoor_suburban	0.7451 $\pm$ 0.0193	0.2703 $\pm$ 0.0272	0.3382 $\pm$ 0.0329	0.4221 $\pm$ 0.0313	0.4752 $\pm$ 0.0236

All pairwise AERIS-versus-baseline comparisons in Table 4 are significant after Holm correction ($p < 0.001$).

At 100 nodes, AERIS ranks first in all four tested environments within the evaluated baseline set (LEACH, PEGASIS, HEED, and TEEN). This claim is restricted to the 100-node matrix and should not

be generalized to the 1000-node regime. Because Table 4 is generated from the legacy 100-node regime (without strict collision/relay flags), its absolute values are intentionally not pooled with the primary large-scale matrix. Figure 2 visualizes the cross-environment PDR ordering at this scale.

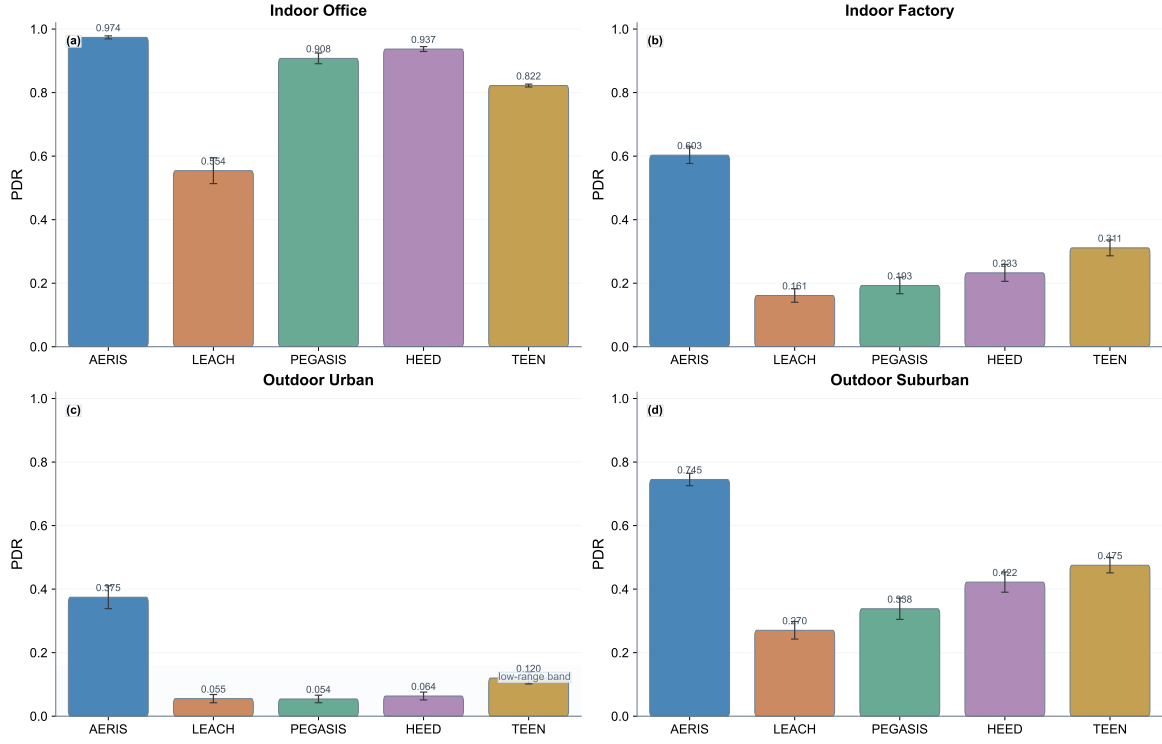


Figure 2. PDR comparison at 100 nodes (4 environments, $n=30$). Shared 0–1 y-axis across panels; outdoor_urban inset highlights the low-range baseline band (0–0.16).

5.2. Ablation: Environment-Dependent Effects ($n=30$)

Table 5. Full vs no_gateway PDR (mean $\pm$ std, reported at 4-decimal precision).

Environment	Full	no_gateway	Delta (no_gateway - full)
indoor_office	0.9739 $\pm$ 0.0047	0.9741 $\pm$ 0.0036	+0.0002
indoor_factory	0.6031 $\pm$ 0.0258	0.5806 $\pm$ 0.0215	-0.0225
outdoor_urban	0.3745 $\pm$ 0.0354	0.3534 $\pm$ 0.0301	-0.0212
outdoor_suburban	0.7451 $\pm$ 0.0193	0.7306 $\pm$ 0.0264	-0.0146

Gateway is positive in three environments and near-neutral in indoor_office (Table 5, Figure 3). CAS is not consistently positive across environments. In indoor_office, the Full-vs-no_gateway gap is +0.0002 (absolute PDR), so this cell is treated as practically neutral. After Holm–Bonferroni correction ($m = 8$), only two comparisons remain significant: Gateway in indoor_factory ($\Delta = -2.25$ pp, $p_{\text{Holm}} = 0.005$) and CAS in outdoor_urban ($\Delta = +2.44$ pp, $p_{\text{Holm}} = 0.015$); see Panel (c) of Figure 3.

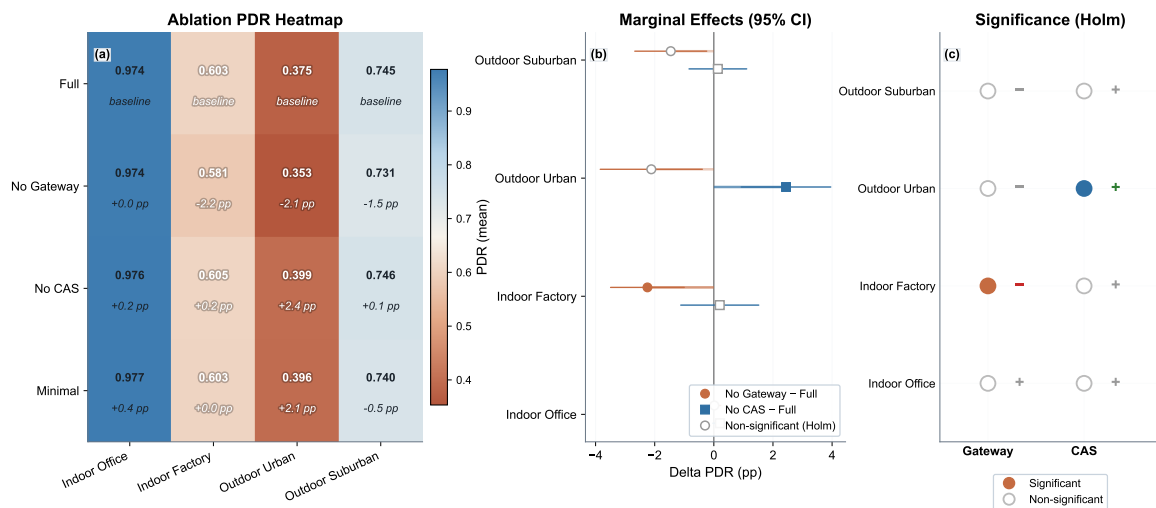


Figure 3. Ablation analysis ($n=30$ per cell, four environments): (a) PDR heatmap with per-cell delta relative to Full; (b) marginal effects of removing Gateway or CAS with 95% Welch confidence intervals (filled markers denote Holm-significant comparisons at $\alpha = 0.05$); (c) Holm-corrected significance overview with effect direction.

5.3. Scalability (100–1000 Nodes; Balanced $n=3200$ per Cell)

Table 6. PDR ranking at 1000 nodes from the primary large-scale matrix ($n=3200$ per cell). Values are means; per-cell SDs range from 0.001 to 0.05 and are reported in the significance heatmap (Figure 5).

Environment	AERIS	LEACH	PEGASIS	HEED	TEEN	AERIS rank
indoor_office	0.6771	0.0282	0.9884	0.0134	0.0170	2nd
indoor_factory	0.7284	0.0108	0.2999	0.0054	0.0070	1st
outdoor_urban	0.1359	0.0041	0.0987	0.0016	0.0025	1st
outdoor_suburban	0.7275	0.0160	0.4725	0.0088	0.0106	1st

AERIS ranks first in 18 of 24 environment-scale cells; the remaining 6 cells (all indoor_office) are won by PEGASIS (Table 6). Unlike the legacy 100-node comparability matrix, the primary large-scale matrix shows physically plausible scale behavior: AERIS PDR decreases from 100 to 1000 nodes in all four environments under explicit collision and relay constraints.

5.4. Statistical Robustness Snapshot (AERIS vs Strongest Baseline)

Table 7. Holm-corrected significance snapshot at 1000 nodes (AERIS vs strongest baseline in each environment; $n=3200$ per cell).

Environment	Nodes	Δ PDR	Hedges' g	Holm p	Significant
indoor_office	1000	-0.311272 (vs PEGASIS)	-14.2041	$< 10^{-300}$	yes
indoor_factory	1000	+0.428522 (vs PEGASIS)	+5.0292	$< 10^{-300}$	yes
outdoor_urban	1000	+0.037265 (vs PEGASIS)	+0.5592	8.43×10^{-107}	yes
outdoor_suburban	1000	+0.255009 (vs PEGASIS)	+2.8235	$< 10^{-300}$	yes

This snapshot shows positive AERIS deltas in three harsh environments and a negative delta in indoor_office, where PEGASIS is the strongest baseline at large scale. For very large sample sizes, practical interpretation in this study still follows both effect size and absolute PDR gap rather than significance labels alone. The very large absolute values of Hedges' g in harsh environments reflect both large mean gaps and low within-group variance under large sample sizes; they should be interpreted

as magnitude indicators within this experiment design, not as cross-study constants.¹ Figure 4 shows the corresponding scalability trends across all five protocols.

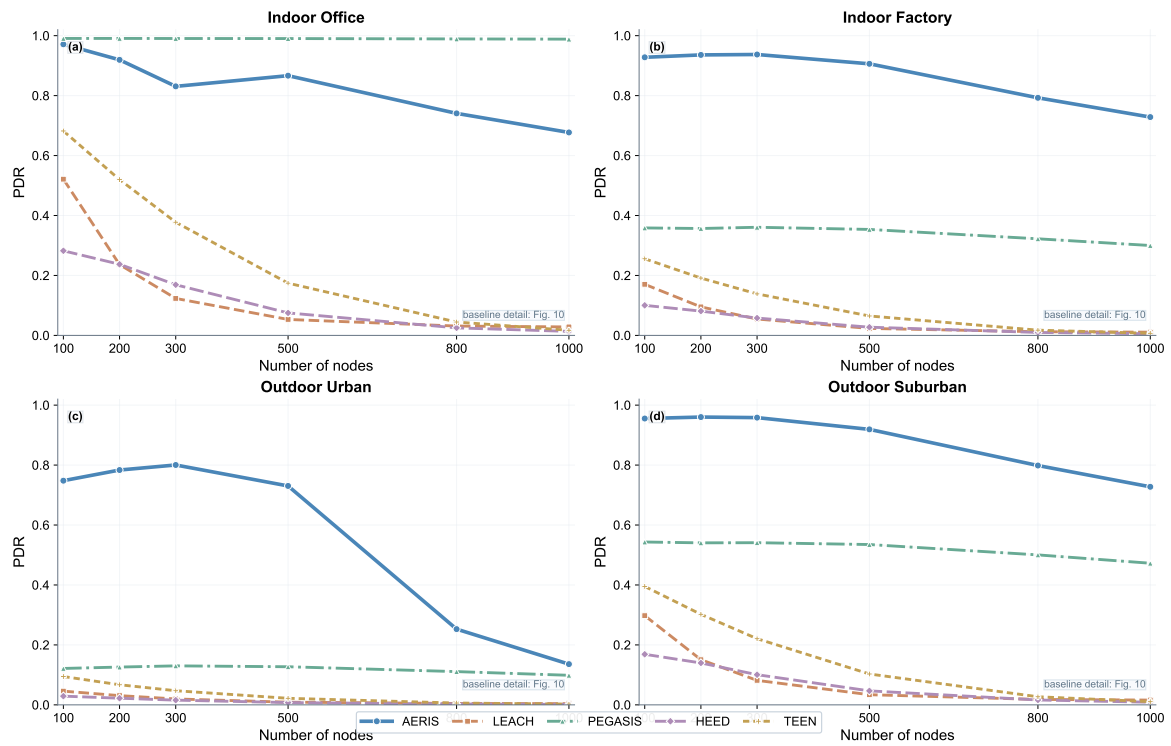


Figure 4. Primary large-scale scalability trends ($n=3200$ per cell, 95% CI bands). Each panel shows all five protocols over six node counts on a shared 0–1 y-axis.

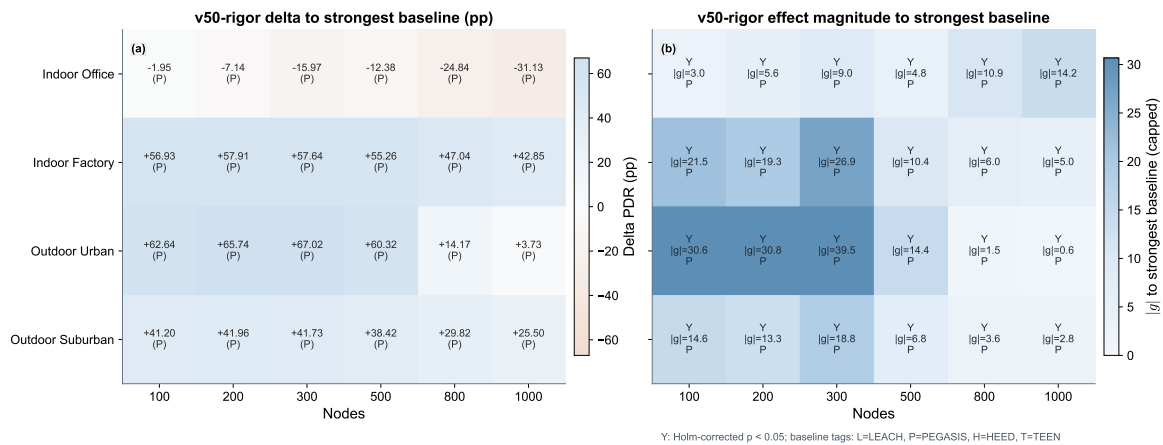


Figure 5. Primary-matrix strongest-baseline map. (a) Cell-wise ΔPDR to the strongest baseline; tags L/P/H/T denote LEACH/PEGASIS/HEED/TEEN. (b) Effect-size magnitude $|g|$ with Holm significance; hatched cells are non-significant.

¹ For large- n comparisons with very small within-group variance, standardized effect sizes can become extremely large. In this manuscript, Hedges' g is therefore interpreted jointly with absolute ΔPDR and confidence intervals, not as a stand-alone cross-study scale.

Figure 5 encodes three quantities per cell: the baseline tag (L/P/H/T), signed ΔPDR , and Holm significance with $|g|$. This makes delta and effect-size statements in Table 7 fully traceable to a concrete baseline protocol.

5.5. Rigor-Patch Pilot Sanity Check ($n=60$)

To test simulator-physics consistency before the full rerun, we ran a local pilot with collision and baseline multi-hop relay enabled, using the same environment taxonomy and nodes {100, 500, 1000} with $n=60$ per cell. This pilot is retained as a diagnostic snapshot (Table A1 in Appendix A) and is not used in primary claim tables.

The key pilot observation is monotonicity recovery: AERIS PDR decreases with scale in all four environments. In indoor_office, PEGASIS is higher than AERIS under this stricter setting, while AERIS remains higher in the other three environments. The pilot therefore served as a gate check for launching the full primary large-scale matrix and is reported only for diagnostic completeness.

5.6. Stress-Delta Matrix A (Four Environments)

To reduce seed and sampling confounders, we ran Stress-delta matrix A (patch-minus-control comparison) in all four environments (indoor_office, indoor_factory, outdoor_urban, and outdoor_suburban). The patch mode enables collision and baseline relay mechanisms; the control mode uses the legacy control configuration. Patch cells use $n=1000$ and control cells use $n=600$ (Table A2 in Appendix A).

Three observations are robust in this stress-delta matrix. First, AERIS remains above LEACH/HEED/TEEN under patch mode in all four environments at 1000 nodes. Second, PEGASIS is minimally affected by the patch, with near-zero deltas and non-significant tests in all 24 PEGASIS cells². A technical hypothesis is that serialized chain forwarding in PEGASIS exposes fewer branch decisions to stress toggles than cluster-centric protocols, so fewer control points are perturbed under the current implementation path. Third, AERIS patch-minus-control deltas are significant in all 24 AERIS cells after Holm correction, with the largest absolute drop in outdoor_urban at high scale. Table A3 (Appendix A) provides the 1000-node PEGASIS snapshot supporting this interpretation.

In matched stress-delta matrix B, PEGASIS shows exact zero deltas in indoor_factory (all six node counts). We treat this as an implementation-coupling anomaly requiring dedicated implementation audit, not as physical invariance evidence. The patch can also be overly aggressive for baseline paths at high scale, so this block is interpreted as stress-delta evidence rather than as a final replacement matrix. To remove sample-size asymmetry from stress-delta matrix A, we run matched stress-delta matrix B (patch $n = 1000$, control $n = 1000$ per cell) and report it separately below.

Consistency between stress-delta matrix A and matched stress-delta matrix B is assessed from the matched-delta tables and significance files; both matrices preserve the same direction pattern for AERIS across environment-scale cells.

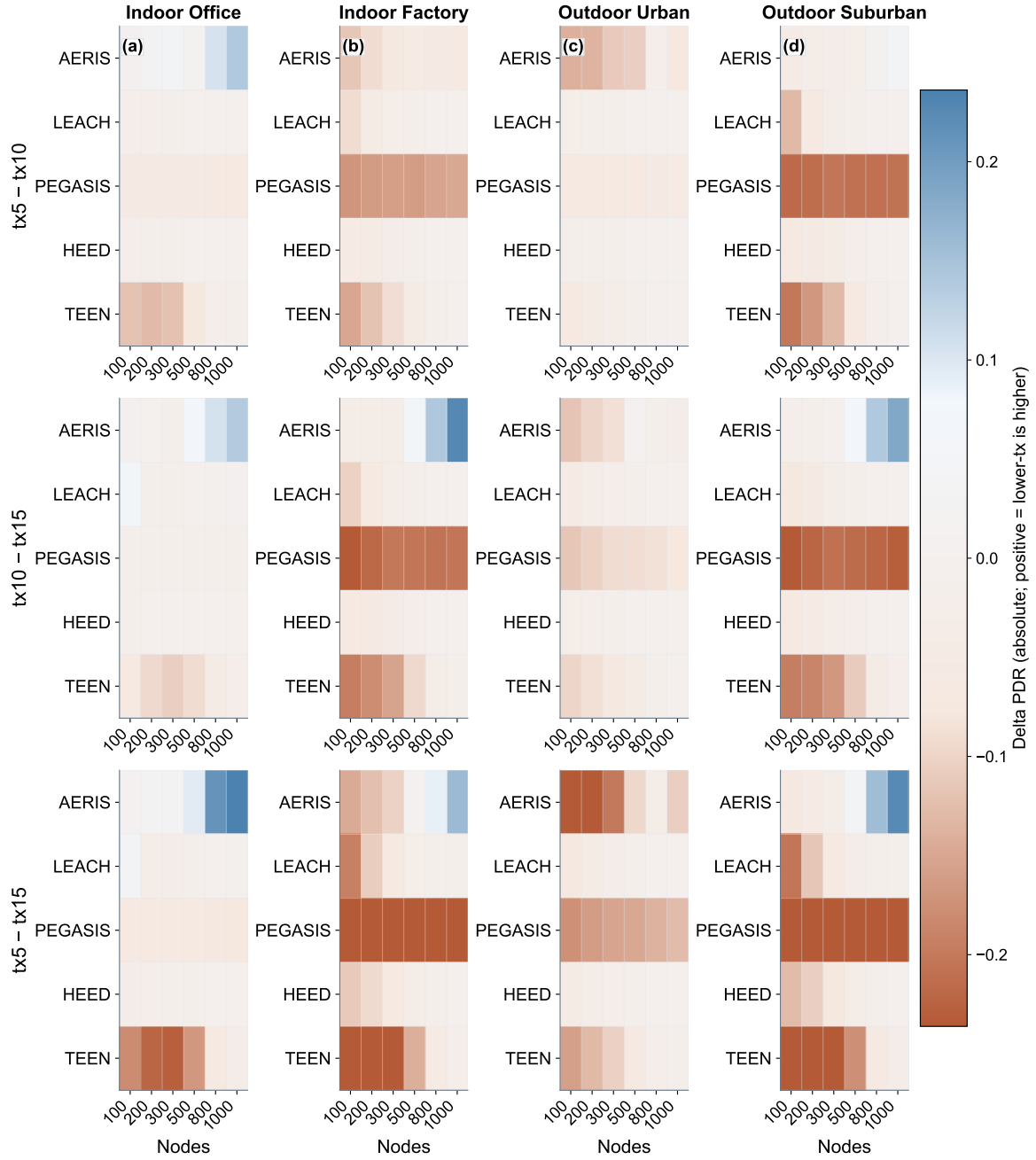
5.7. TX-Power Sensitivity Block (Four Environments)

We additionally evaluate tx-power sensitivity under the upgraded simulation path by comparing 5 dBm, 10 dBm, and 15 dBm across four environments, six node counts (100, 200, 300, 500, 800, 1000), and five protocols ($n=1000$ per cell). The full pairwise matrix contains 360 matched comparisons (4 environments $\times$ 6 node counts $\times$ 5 protocols $\times$ 3 pairwise tx comparisons), and all 360 are significant after Holm correction.

² Across the 24 PEGASIS cells in this stress-delta matrix, $|\Delta_{\text{stress-delta}}| \leq 0.0044$, all Holm-adjusted p -values are 1.0000, and $\max |g| = 0.0903$, consistent with a near-zero effect.

Table 8. AERIS power-sensitivity snapshot at 1000 nodes (mean PDR, tx5/tx10/tx15) from the power-sensitivity matrix (n=1000 per cell).

Environment	tx=5 dBm	tx=10 dBm	tx=15 dBm	Delta (5–10)	Delta (5–15)
indoor_office	0.8176	0.6775	0.5404	+0.1401	+0.2771
indoor_factory	0.6633	0.7291	0.5032	-0.0658	+0.1601
outdoor_urban	0.0582	0.1363	0.1634	-0.0781	-0.1052
outdoor_suburban	0.7679	0.7269	0.5431	+0.0411	+0.2249

**Figure 6.** Power-sensitivity delta maps in absolute PDR units (rows: tx5–tx10, tx10–tx15, tx5–tx15; columns: four environments; five protocols, six node counts). Cross markers denote Holm non-significant cells.

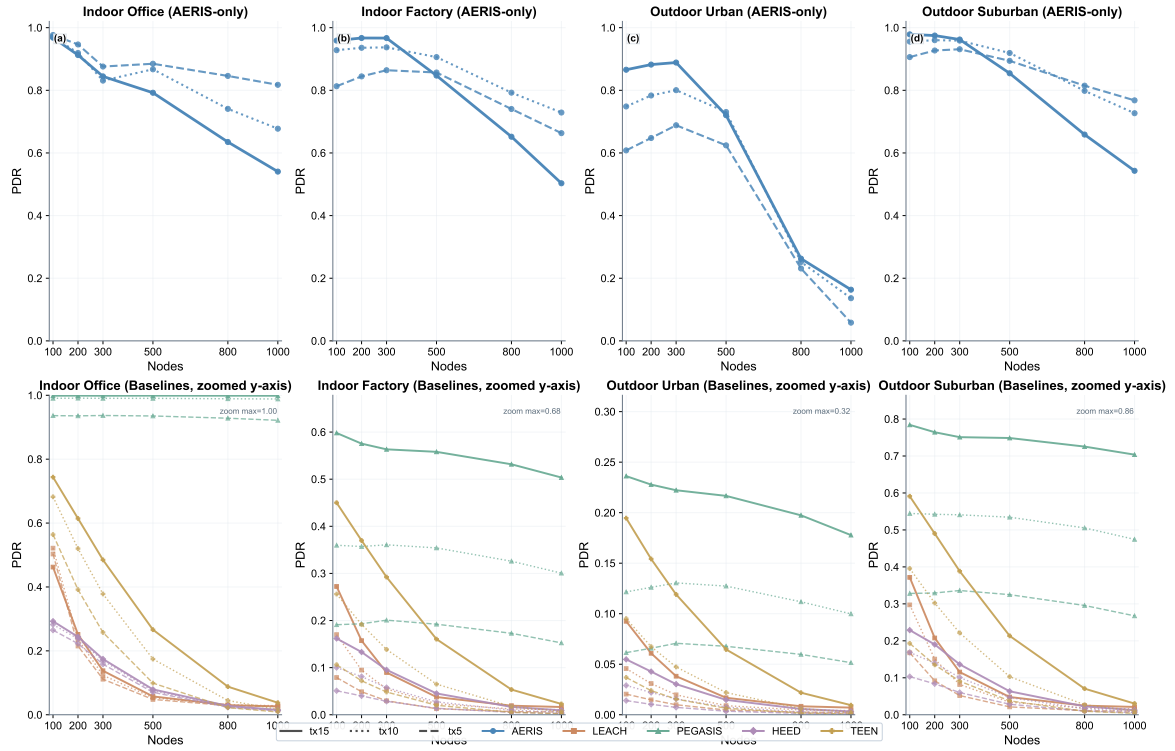


Figure 7. Absolute PDR profiles from the power-sensitivity matrix. Top row: AERIS (line style = tx level, common y-axis 0–1). Bottom row: baseline protocols with environment-specific y-axis zoom for low-PDR readability.

The tx-power block (Table 8, Figures 6 and 7) confirms that response is environment-dependent and non-monotonic under the current routing and channel settings. In particular, raising tx-power does not guarantee higher PDR in every environment-node cell. A plausible mechanism is interference-domain expansion: higher transmit power increases both reachable receivers and competing transmitters, which can increase collision probability in dense regimes and partially offset link-budget gains. In the completed power-sensitivity matrix, all 360 pairwise tx comparisons are significant after Holm correction. Two targeted cells are highlighted because they are easy to misread under a monotonicity assumption. First, indoor_factory at 500 nodes follows $tx_{10} > tx_5 > tx_{15}$ for AERIS (0.9060, 0.8569, 0.8471), so the local optimum is mid-power rather than high-power. Second, outdoor_urban at 800 nodes shows high dispersion for AERIS ($CV_{tx_5} = 0.54$, $CV_{tx_{10}} = 0.44$, $CV_{tx_{15}} = 0.34$), indicating a near-threshold regime where connectivity and collision states alternate across replicates. This variance concentration at high scale is consistent with intermittent percolation behavior in sparse urban connectivity, where small contention shifts can move rounds between connected and fragmented states.

5.8. Matched Stress-Delta Confirmation Matrix (Four Environments, $n=1000$ vs 1000)

To remove remaining sample-size asymmetry from stress-delta matrix A, we completed matched stress-delta matrix B in which both patch and control arms use $n=1000$ per environment-node-protocol cell (4 environments $\times$ 6 node counts $\times$ 5 protocols). This matrix directly tests whether stricter physics assumptions improve or degrade reliability under matched sampling (Table A4 in Appendix A).

In the matched matrix, AERIS patch-minus-control deltas are negative and significant in all 24 environment-scale cells after Holm correction (24/24 significant; Hedges' g range: -0.47 to -15.36). Therefore, under the current patch configuration (MAC collision and multi-hop relay enabled), stricter physics assumptions reduce reliability rather than increase it. This matrix is interpreted as realism-stress evidence, not as a replacement for the primary ranking matrix.

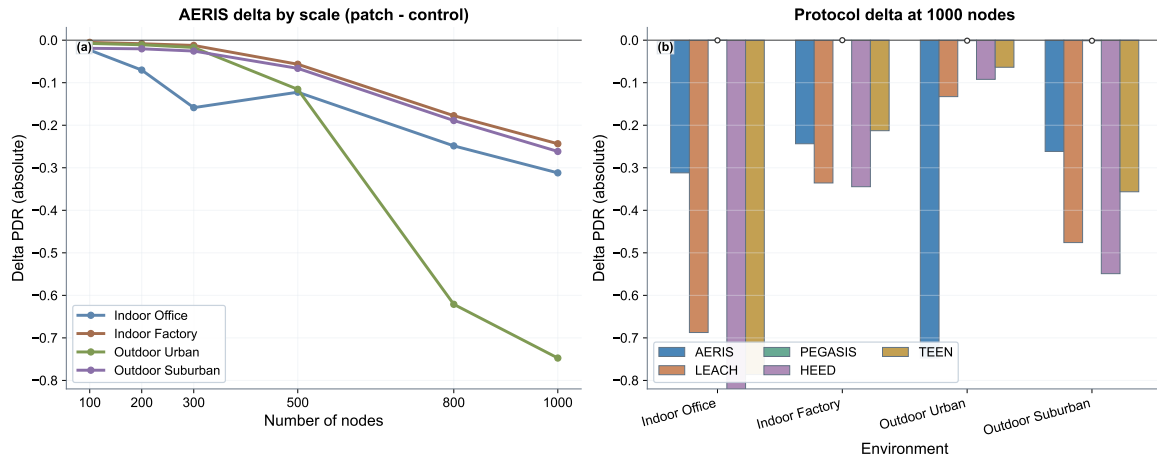


Figure 8. Matched stress-delta evidence. (a) AERIS delta trajectories over node counts (patch - control). (b) Protocol deltas at 1000 nodes across environments. Negative deltas indicate lower PDR under patch mode.

At the protocol level, Holm-significant cell counts in matched stress-delta matrix B are 24/24 for AERIS, 24/24 for LEACH, 0/24 for PEGASIS, 24/24 for HEED, and 24/24 for TEEN. Together with Figure 8, this confirms that the strict patch strongly affects most protocol families, while PEGASIS remains nearly invariant in this matched-delta block. A working technical hypothesis is that the current patch primarily perturbs cluster-based decision points (collision-sensitive parallel uplinks and relay augmentation), whereas PEGASIS uses a serialized chain-forwarding pattern with fewer concurrent uplink competitions in this simulator path; this hypothesis remains subject to explicit code-path instrumentation.

5.9. Hop-Based Latency and Trade-offs

At 100 nodes ($n=30$), AERIS stays near two hops (1.97–1.99), while PEGASIS remains above 31 hops due to chain traversal. LEACH and TEEN show lower average hops because of protocol paths that allow direct-to-BS transmissions.

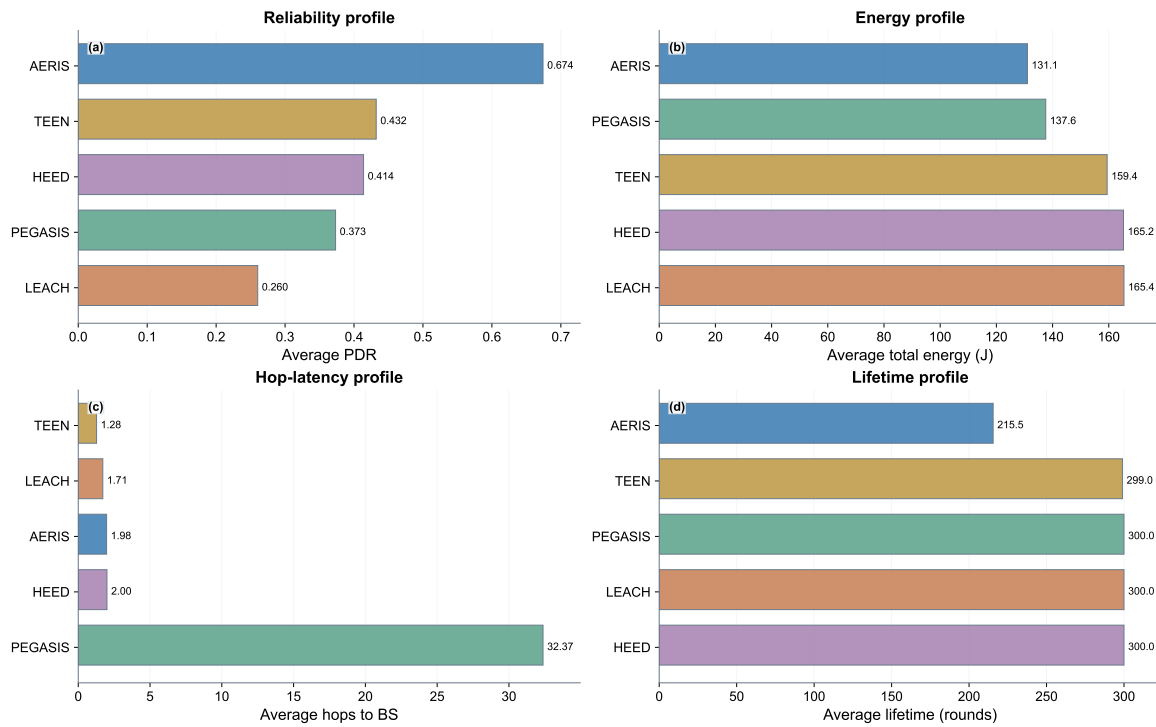


Figure 9. Trade-off panel from publication-tier data: reliability (PDR), energy (J), hop-based latency (average hops), and lifetime (rounds). The hop panel uses a logarithmic x-axis to keep low-hop and high-hop protocols visible in the same frame without overlap.

Figure 9 is used as the consolidated reliability-energy-latency-lifetime trade-off view.

Table 9. Energy-focused statistical snapshot from the publication 100-node matrix (pooled across four environments, $n=120$ per protocol). Delta energy is reported as baseline minus AERIS; positive values indicate higher energy consumption than AERIS.

Protocol	Avg. energy (J)	Avg. lifetime (rounds)	ΔE vs AERIS (J)	Holm-corrected p (energy)
AERIS	131.1	215.5	–	–
LEACH	165.4	300.0	+34.4	5.27×10^{-40}
PEGASIS	137.6	300.0	+6.5	3.99×10^{-4}
HEED	165.2	300.0	+34.2	5.13×10^{-40}
TEEN	159.4	299.0	+28.4	1.34×10^{-32}

Table 9 shows that AERIS consumes the least total energy among the five protocols, yet exhibits the shortest network lifetime (215 rounds vs 300 for PEGASIS/LEACH/HEED). The mechanism is relay-load concentration: Gateway and Skeleton forwarding paths improve delivery but funnel traffic through structurally important nodes, causing earlier first-node death than chain-rotation protocols that distribute load more uniformly. For deployment, this means reliability-first operation under AERIS should be paired with environment-tuned duty cycling or node redundancy to mitigate relay bottleneck depletion.

5.10. NS-3 Trend Validation (AERIS vs LEACH, $n=30$)

NS-3 validation is used as a cross-platform trend check rather than a numerical-equivalence proof. Formal significance claims in this block remain restricted to AERIS-versus-LEACH. Under aligned multi-environment settings, AERIS is directionally not lower than LEACH in 27 of 28 tested environment-scale cells; the single exception is indoor_office at 200 nodes (AERIS–LEACH diff = -0.0004 , not significant). At 100 nodes, significance is confirmed in 3/4 environments

(indoor_factory, outdoor_urban, outdoor_suburban), while indoor_office remains non-significant. In the 300/500-node extension, all eight environment-scale comparisons are significant after Holm correction; with the 800/1000 extension included, non-significance remains confined to indoor_office at 1000 nodes. Table 10 reports the 100-node baseline, and Figure 10 overlays PEGASIS/HEED/TEEN curves for descriptive context only.

Table 10. NS-3 trend-level validation at 100 nodes (AERIS vs LEACH, $n=30$; metric: mean PDR). Per-cell SDs range from 0.003 to 0.04; full distributions are available in the data repository.

Environment	AERIS	LEACH	Delta	Holm-corrected p	Significance
indoor_office	0.9202	0.9185	+0.0017	1.00	no
indoor_factory	0.6025	0.5336	+0.0690	4.23×10^{-25}	yes
outdoor_urban	0.2064	0.1899	+0.0165	3.41×10^{-3}	yes
outdoor_suburban	0.7771	0.6921	+0.0850	1.94×10^{-30}	yes

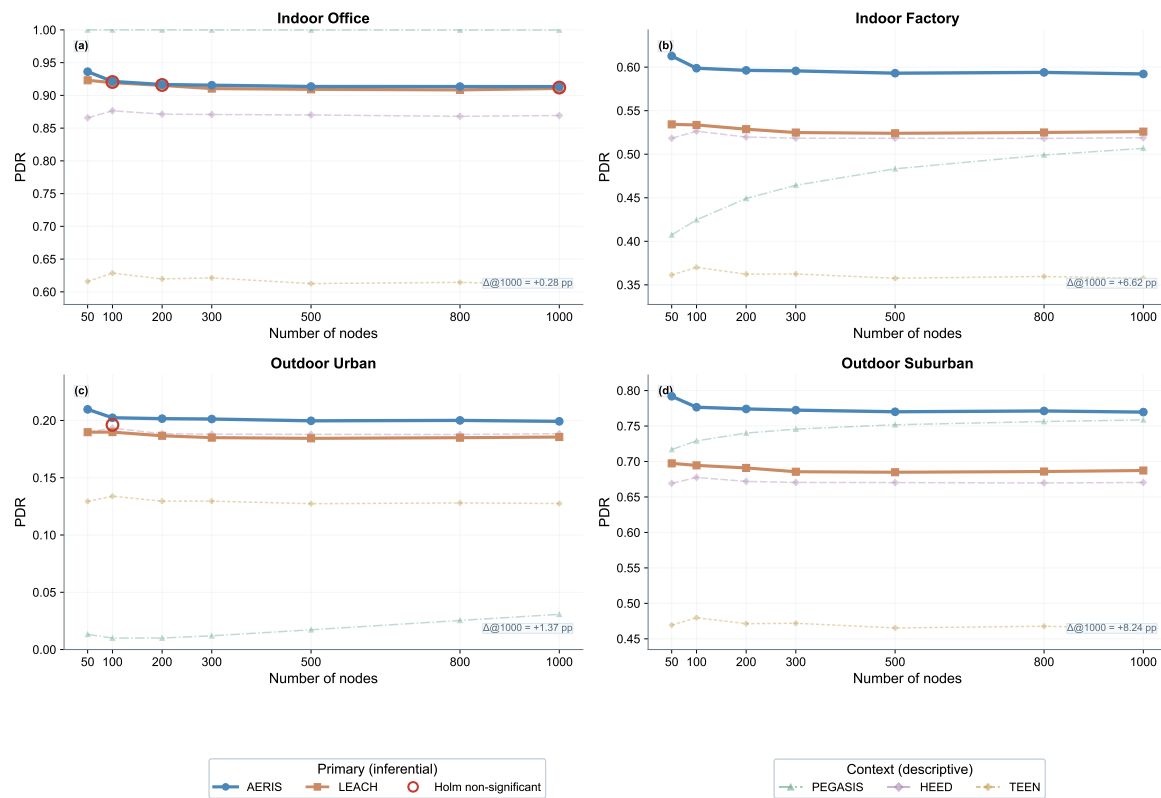


Figure 10. NS-3 trend validation (50–1000 nodes). Primary inferential pair: AERIS vs LEACH (bold lines, significance markers). PEGASIS/HEED/TEEN are overlaid as descriptive context only (faded lines). Red hollow circles mark Holm non-significant scales for AERIS-vs-LEACH.

In this NS-3 block, inferential interpretation is restricted to the AERIS-versus-LEACH pair; context overlays are descriptive only. This block supports trend consistency only; cross-platform numerical equivalence is not claimed. Across 7 node counts and 4 environments, 25/28 AERIS-versus-LEACH comparisons are significant; the non-significant cells are indoor_office at node counts 100, 200, and 1000. Figure 10 visualizes the full 50–1000 node NS-3 trend envelope and the associated significance pattern.

6. Discussion

The evidence supports environment-scoped positioning rather than universal superiority. In the reported matrices, AERIS is strongest on reliability in harsh channels.

The manuscript separates three evidence blocks—100-node comparison, primary large-scale matrix ($n=3200$ per cell), and NS-3 trend validation—without cross-block numerical pooling, because each block answers a different question. Under stricter stress-delta settings, high-scale PDR decreases for most protocols; under tx-power sensitivity, power effects are strong but not globally monotonic. Practical interpretation therefore remains environment-scoped.

An additional boundary is causal interpretation across regime tiers. The difference between the legacy 100-node matrix and the primary large-scale matrix is not interpreted as a single-factor treatment effect, because two changes are introduced together in the primary rerun: (i) strict collision-and-relay physics is enabled, and (ii) previously identified baseline suppressors are removed. For this reason, cross-tier comparisons are used only for scope positioning, while matched stress-delta blocks are used for factor-specific direction checks.

Compared with recent adaptive-routing studies [12,13,15,16,19,24–26,28,30], the main distinction is methodological: rather than reporting a single favorable matrix, this manuscript reports primary ranking evidence alongside matched stress/sensitivity blocks that expose when absolute reliability degrades. A secondary distinction is architectural: AERIS uses fixed-coefficient linear scoring (no training loop, no gradient updates, no policy network), so its inference cost is $O(K)$ per cluster per round, where K is the number of candidate modes. This makes AERIS computationally lightweight and deployment-oriented, avoiding the memory, convergence, and non-stationarity risks associated with online learning. Direct numerical comparison with RL-based protocols is deferred to future work because fair comparison requires aligned reward definitions and convergence criteria that are not standardized across the cited studies.

Practical interpretation also requires separating statistical and engineering significance. At large sample sizes, very small absolute differences can be significant after correction. For this reason, we interpret each result jointly by corrected p value, effect size, and absolute PDR delta. Energy interpretation is similarly conditioned: lower total energy can co-occur with shorter lifetime when earlier network termination occurs.

Interpretation of Matched Degradation Block.

Enabling the current patch configuration (MAC-collision plus multihop relay) reduces absolute AERIS PDR in all 24 matched cells, while relative ranking is preserved. This does not contradict the primary ranking result: ranking asks whether AERIS remains higher than alternatives under the same strict setting, while the matched stress-delta matrix asks how far the strict patch shifts each protocol relative to its own control path. Strict-physics penalties affect all protocols but non-uniformly across protocol families. The correct interpretation is therefore “AERIS keeps relative advantages in harsh regimes while absolute reliability drops under stricter assumptions.”

Practical Deployment Guidance.

Based on the completed matrices, the following deployment guidance is supported by the reported evidence:

- In harsh environments (indoor_factory, outdoor_urban, outdoor_suburban), enabling Gateway support is recommended because no-gateway ablations reduce reliability.
- In benign indoor_office conditions, gains are smaller and model overhead should be justified against application reliability targets.
- For high-scale deployments, use the primary large-scale matrix as primary evidence and use stress/sensitivity blocks for calibration.
- Tx-power settings should be environment-tuned; monotonic power assumptions are not supported by the sensitivity block (see Section 5.5 for specific counter-examples).

Table 11. Environment-scoped engineering interpretation from the reported evidence.

Environment type	Recommended interpretation	Main caveat
indoor_office	Reliability gaps are small at high scale; deployment cost and simplicity can dominate design choice	Some NS-3 cells are non-significant
indoor_factory	AERIS with Gateway support gives the most stable reliability gain	Calibrate tx-power and collision parameters
outdoor_urban	AERIS remains preferable under harsh links in both simulator regimes	High-scale behavior is sensitive to physics settings
outdoor_suburban	AERIS trend remains favorable in simulator and NS-3 trend checks	Do not assume monotonic power-response

Validity Notes.

Three boundaries are explicit in this draft (see Table 11 for environment-scoped guidance). First, the reported comparisons are simulator-based under the tested channel taxonomy. Second, NS-3 evidence is used for cross-platform trend validation (AERIS vs LEACH), not full five-protocol numerical alignment. Third, module contributions are context dependent: Gateway and CAS effects vary by environment and are not globally monotonic.

7. Limitations and Future Work

- Current evidence is simulation-based under the tested channel taxonomy.
- NS-3 validation in this manuscript focuses on AERIS versus LEACH; full five-protocol cross-platform ranking is future work.
- The primary large-scale matrix ($n=3200$ per cell) includes explicit collision and relay constraints, but does not replace hardware-in-the-loop validation. A priority next step is deploying AERIS on a TelosB or nRF52-based testbed (10–50 nodes) to confirm that the simulated environment-dependent PDR ordering holds under real radio impairments including multipath fading, clock drift, and co-channel interference from non-WSN devices.
- Stress-delta matrix A (patch minus control), the tx-power sensitivity matrix, and the matched stress-delta matrix B show that stricter physics assumptions and tx-power changes can materially alter absolute PDR, so deployment requires environment-specific calibration.
- In the matched stress-delta matrix, PEGASIS shows near-invariant patch-minus-control deltas; this behavior is treated as a simulator-coupling artifact requiring dedicated implementation audit. A working technical hypothesis is that serialized chain forwarding exposes fewer branch points to patch toggles than cluster-centric baselines under the current code path. In the primary large-scale ranking matrix, all protocols (including PEGASIS) are executed under the same strict collision-and-relay path without mixing a control arm, so this artifact affects interpretation of stress-delta blocks but does not alter standalone primary-matrix ranking values, which are computed under one strict path without control-arm subtraction.
- Reported gains are limited to the tested baseline set and topology families in this manuscript.
- Hop count is a route-length proxy, not a wall-clock latency benchmark under full MAC scheduling.
- In high-CV cells (e.g., outdoor_urban at 800 nodes, where CV exceeds 0.33 across all three tx-power levels), $\text{mean} \pm \text{std}$ may understate distributional complexity; median-based or quantile summaries are recommended for deployment-critical decisions in such near-threshold regimes.

8. Conclusion

AERIS is a reliability-oriented routing option for heterogeneous WSN channels. In the legacy 100-node comparability matrix, it achieves the highest mean PDR within the tested baseline set across all four environments. In the primary large-scale matrix (100–1000 nodes, four environments), AERIS

is rank-1 in three environments (indoor_factory, outdoor_urban, outdoor_suburban) and rank-2 in indoor_office where PEGASIS remains dominant.

NS-3 provides trend-level cross-platform support for AERIS versus LEACH. The completed stress and sensitivity blocks show that stricter physics assumptions and tx-power changes materially affect high-scale behavior. In the matched stress-delta matrix, AERIS patch-minus-control deltas are negative and significant in all 24 environment-scale cells under the current patch setup.

The practical conclusion is environment-scoped deployment: AERIS remains strong for reliability-first operation in harsh channels, while benign indoor conditions at large scale may favor chain-forwarding baselines. Final parameterization should therefore be environment-tuned using matched stress and sensitivity evidence.

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Data Availability Statement

The datasets, scripts, manuscript sources, figure-generation code, and provenance sidecar records (including commit identifiers and script hashes) supporting this study are available from the corresponding author and will be released in a versioned public repository upon final publication. The released archive includes the full 100-node benchmark matrix, the primary large-scale matrix and significance tables, stress/sensitivity calibration matrices, and the NS-3 trend-level validation summary.

Appendix A Diagnostic and Stress-Delta Tables

The following tables are referenced in the main text for regime-transition diagnostics and stress calibration. They are not used for primary ranking claims.

Table A1. AERIS PDR under the rigor-patch pilot (n=60 per environment-node cell). This pilot table is included for regime-transition diagnostics and is not used for primary claims.

Environment	100 nodes	500 nodes	1000 nodes
indoor_office	0.9706	0.8676	0.6746
indoor_factory	0.9283	0.9083	0.7355
outdoor_urban	0.7487	0.7362	0.1492
outdoor_suburban	0.9563	0.9193	0.7307

Table A2. AERIS PDR in stress-delta matrix A (patch-minus-control comparison, four environments).

Environment	Nodes	Patch	Control
indoor_office	100	0.9714	0.9948
indoor_office	500	0.8679	0.9902
indoor_office	1000	0.6779	0.9899
indoor_factory	100	0.9281	0.9329
indoor_factory	500	0.9060	0.9627
indoor_factory	1000	0.7291	0.9726
outdoor_urban	100	0.7482	0.7559
outdoor_urban	500	0.7299	0.8460
outdoor_urban	1000	0.1372	0.8845
outdoor_suburban	100	0.9556	0.9744
outdoor_suburban	500	0.9198	0.9859
outdoor_suburban	1000	0.7280	0.9897

Table A3. PEGASIS stress-delta snapshot at 1000 nodes from stress-delta matrix A (n=1000 patch, n=600 control).

Environment	Delta	Hedges' g	Holm p	Significant
indoor_office	-0.0003	-0.0435	1.0000	No
indoor_factory	+0.0006	+0.0057	1.0000	No
outdoor_urban	+0.0017	+0.0252	1.0000	No
outdoor_suburban	-0.0011	-0.0085	1.0000	No

Table A4. AERIS patch-minus-control deltas in the matched stress-delta matrix (delta = patch - control).

Environment	Delta at 100 nodes	Delta at 1000 nodes
indoor_office	-0.0234	-0.3120
indoor_factory	-0.0052	-0.2435
outdoor_urban	-0.0077	-0.7474
outdoor_suburban	-0.0191	-0.2616

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